

NATIONAL UNIVERSITY OF SINGAPORE

CS5275 - The Algorithm Designer's Toolkit

(Semester 2 : AY2024/2025)

Time allowed : 2 hours

Student Number: _____

INSTRUCTIONS TO CANDIDATES

1. Write your matriculation number clearly in the space above. Do not write your name.
2. This paper contains **NINE (9)** questions and comprises **FOURTEEN (14)** printed pages (including this cover page, and 3 extra pages at the end for overflow answers). The total score is 80.
3. Do **NOT** answer all questions – instead please follow the instructions on **Page 2** and indicate which questions you choose to answer.
4. This is an **OPEN BOOK** assessment. Calculators, phones, electronic devices, and Internet access are **NOT** permitted.
5. Write your answers in the spaces provided. **Further space is available at the back** if needed, but it must be made clear which answers are continued there, and all answers there must be **labeled clearly** with the exact question number.

Examiner's Use Only	
Questions	Marks
1a	
1b	
1c	
2a	
2b	
2c	
3a	
3b	
3c	
Total	

Required: Indication of Questions Answered

The marking scheme is as follows (80 marks total):

- Core part (72 Marks): Choose **two questions** to answer among Q1a,Q1b,Q1c, two among Q2a,Q2b,Q2c, and two among Q3a,Q3b,Q3c. These will comprise **12 marks each**.
- Semi-optional part (8 Marks): Among the **three questions not answered in the core part**, you may choose to answer **up to two of them** for **4 marks each**.

Please clearly indicate your choices below. If your choices are unclear, incomplete, or inconsistent, then they may be replaced by choices made solely based on the length of each answer (e.g., the two longest answers being assigned 'core').

Group 1: Tick **two questions** among the following for the **core part** (12 marks each):

Q1a (Concentration)	Q1b (MWU)	Q1c (E.C. Codes)

Group 2: Tick **two questions** among the following for the **core part** (12 marks each):

Q2a (Prob. Method)	Q2b (Convex Opt.)	Q2c (Fourier)

Group 3: Tick **two questions** among the following for the **core part** (12 marks each):

Q3a (Submod./I.T.)	Q3b (Expanders)	Q3c (Comm. Complexity)

Semi-optional part: Among the 3 questions that you didn't tick above, enter up to two that you will answer for 4 marks each. The remaining one will not be worth any marks.

Semi-optional questions answered

Question 1a: Concentration Inequalities

(Reminder: Do not answer all questions. Each question may be worth 12 Marks (Core Part), 4 Marks (Semi-Optional Part), or 0 Marks, and you must clearly indicate your choices on Page 2.)

Suppose that a randomized algorithm outputs the correct answer with probability $\frac{1}{2} + \epsilon$ for some $\epsilon \in (0, \frac{1}{4})$. Suppose that the goal is to boost the success probability to $1 - \delta$ for some parameter $\delta \in (0, \frac{1}{4})$ by independently running the algorithm n times and combining the results.

Give the details of how the n outputs should be combined, and use Hoeffding's inequality to show that the above goal can be achieved when $n \geq n^*(\delta, \epsilon)$, where $n^*(\delta, \epsilon)$ is some suitable function of δ and ϵ . Show your working, and note that $n^*(\delta, \epsilon)$ should be non-asymptotic (no $O(\cdot)$ notation).

Question 1b: Multiplicative Weight Updates

(Reminder: Do not answer all questions. Each question may be worth 12 Marks (Core Part), 4 Marks (Semi-Optional Part), or 0 Marks, and you must clearly indicate your choices on Page 2.)

Consider the problem of prediction with expert advice, with $n = 2$ experts and $T = 4$ iterations. Suppose that the two experts have the following loss sequences:

- Expert 1: $m_1^{(1)} = 1, m_1^{(2)} = \frac{1}{2}, m_1^{(3)} = 0, m_1^{(4)} = 0$
- Expert 2: $m_2^{(1)} = 0, m_2^{(2)} = \frac{1}{2}, m_2^{(3)} = 1, m_2^{(4)} = 0$

Consider running the Multiplicative Weight Update (MWU) algorithm in which the multiplicative factor is $1 - \eta m_i^{(t)}$, we set $\eta = \frac{1}{2}$, and the weights are initialized to $w_1^{(1)} = w_2^{(1)} = 1$. Derive the algorithm's distribution over experts $\mathbf{p}^{(t)}$ for $t = 1, 2, 3, 4$, showing your working.

Question 1c: Error-Correcting Codes

(Reminder: Do not answer all questions. Each question may be worth 12 Marks (Core Part), 4 Marks (Semi-Optional Part), or 0 Marks, and you must clearly indicate your choices on Page 2.)

Consider the error-correcting code with $k = 2$ and $n = 6$ such that if the input bits are $\mathbf{u} = [u_1, u_2]$, then the resulting codeword is $\mathbf{x} = [u_1, u_1, u_1, u_2, u_2, u_2]$. Find the generator matrix $\mathbf{G}$ and a parity check matrix $\mathbf{H}$ for this code, explaining briefly.

(Note: There are multiple valid answers for $\mathbf{H}$; any single one suffices as long as you explain correctly.)

Question 2a: The Probabilistic Method

(Reminder: Do not answer all questions. Each question may be worth 12 Marks (Core Part), 4 Marks (Semi-Optional Part), or 0 Marks, and you must clearly indicate your choices on Page 2.)

Suppose that n sports teams labeled $\{1, \dots, n\}$ each play a match against one another ($\binom{n}{2}$ matches in total), and for each match, one team wins and the other team loses (no draws). The matches are played one-by-one in some (arbitrary) pre-specified order, creating a sequence of $\binom{n}{2}$ match outcomes.

Fix any $k \in \{1, 2, \dots, n-1\}$. Using the probabilistic method, show that if $\binom{n}{k}(1-2^{-k})^{n-k} < 1$ then there exists a (length- $\binom{n}{2}$) sequence of match outcomes satisfying the following property: *For any subset of teams $S \subseteq \{1, \dots, n\}$ of size k (i.e., $|S| = k$), there exists another team $u \in \{1, \dots, n\} \setminus S$ such that u wins against every team in S .*

Question 2b: Convex Optimization

(Reminder: Do not answer all questions. Each question may be worth 12 Marks (Core Part), 4 Marks (Semi-Optional Part), or 0 Marks, and you must clearly indicate your choices on Page 2.)

Consider the following optimization problem:

$$\text{minimize}_{\mathbf{x} \in \mathbb{R}^d} \|\mathbf{x}\| \quad \text{subject to} \quad \sum_{i=1}^d x_i = 1, \mathbf{x} \geq 0,$$

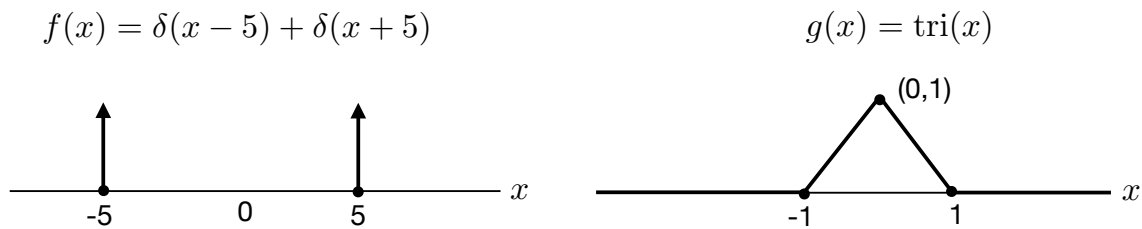
where $\|\cdot\|$ is the usual Euclidean norm, x_i is the i -th entry of $\mathbf{x}$, and the inequality $\mathbf{x} \geq 0$ means every element of $\mathbf{x}$ is non-negative.

Using the method of Lagrange multipliers, find the minimizer $\mathbf{x}^*$. *(Hint: You are allowed to transform the above problem into a different problem, solve that problem using Lagrange multipliers, and then argue that the original problem has the same minimizer. However, significant penalties will be imposed if you don't use Lagrange multipliers and instead find $\mathbf{x}^*$ using other methods.)*

Question 2c: Fourier Transform

(Reminder: Do not answer all questions. Each question may be worth 12 Marks (Core Part), 4 Marks (Semi-Optional Part), or 0 Marks, and you must clearly indicate your choices on Page 2.)

Consider the signals $f(x)$ and $g(x)$ shown in the following figure:



As usual, let $\hat{f}(\omega), \hat{g}(\omega)$ be the corresponding Fourier transforms. Suppose that there are three signals $h_1(x), h_2(x), h_3(x)$ whose Fourier transforms $\hat{h}_1(\omega), \hat{h}_2(\omega), \hat{h}_3(\omega)$ are known to satisfy the following:

- $\hat{h}_1(\omega) = \hat{f}(\omega) + \hat{g}(\omega)$;
- $\hat{h}_2(\omega) = \hat{f}(x)\hat{g}(x)$;
- $\hat{h}_3(\omega) = (\hat{f} * \hat{g})(\omega)$ with $*$ denoting convolution (i.e., $\hat{h}_3(\omega) = \int_{-\infty}^{\infty} \hat{f}(\omega - u)\hat{g}(u)du$).

Draw three separate figures plotting $h_1(x)$, $h_2(x)$, and $h_3(x)$ respectively, explaining (very) briefly. Label any “significant” points in your figures, analogous to the labels 5, -5 , 1, -1 , and $(0, 1)$ above.

Question 3a: Submodularity and Information Theory

(Reminder: Do not answer all questions. Each question may be worth 12 Marks (Core Part), 4 Marks (Semi-Optional Part), or 0 Marks, and you must clearly indicate your choices on Page 2.)

Let $(X_1, \dots, X_n)$ be a collection of random variables (not necessarily independent), and define the following function on subsets $S \subseteq \{1, \dots, n\}$:

$$f(S) = H(X_S) \text{ if } S \neq \emptyset, \quad (\text{and in the case } S = \emptyset \text{ define } f(\emptyset) = 0)$$

where $X_S = \{X_i\}_{i \in S}$ and $H(\cdot)$ denotes entropy. For example, $f(\{1, 3, 5\}) = H(X_1, X_3, X_5)$.

Prove that $f(S)$ is monotone and submodular. *(Hint: Multiple properties of entropy are useful.)*

Question 3b: Expander Graphs

(Reminder: Do not answer all questions. Each question may be worth 12 Marks (Core Part), 4 Marks (Semi-Optional Part), or 0 Marks, and you must clearly indicate your choices on Page 2.)

Consider a bipartite graph G with vertex sets L (with $|L| = n$) and R (with $|R| = m$) such that each node in L is connected to exactly d nodes in R . As a special case of the lecture's definition, we say G is an (n, m, d, ϵ) -bipartite-expander if, for any subset $S \subseteq L$ with $|S| > 0$, the number of neighbors satisfies $|N(S)| \geq \epsilon d|S|$. Let $\epsilon^*(G)$ be the largest ϵ such that G is an (n, m, d, ϵ) bipartite expander.

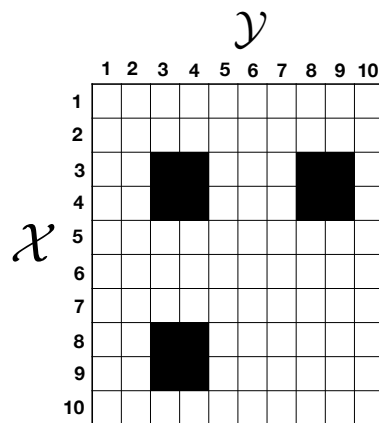
Set $n = 4$, $d = 2$, and $m = 8$. Draw two bipartite graphs that are consistent with these parameter choices, one of which makes $\epsilon^*(G)$ as high as possible, and the other of which makes $\epsilon^*(G)$ as small as possible. State the resulting values of $\epsilon^*(G)$, and explain why they are the highest/smallest possible.

(Note: Since we are considering graphs and not multi-graphs, any two nodes $i \in L$ and $j \in R$ can have at most a single direct edge between them, not multiple.)

Question 3c: Communication Complexity

(*Reminder: Do not answer all questions. Each question may be worth 12 Marks (Core Part), 4 Marks (Semi-Optional Part), or 0 Marks, and you must clearly indicate your choices on Page 2.*)

Consider the function $f(x, y)$ defined on the sets $\mathcal{X} = \mathcal{Y} = \{1, 2, \dots, 10\}$ and depicted in the figure below, where shaded entries represent $f(x, y) = 1$ and non-shaded entries indicate $f(x, y) = 0$.



Answer the following:

- (i) Describe a deterministic protocol that computes $f(x, y)$ using 3 bits of communication. The final (third) bit sent in the protocol should be function value $f(x, y)$ itself.
- (ii) Using one of the lower bounding techniques from the lecture, prove that any deterministic protocol requires at least 3 bits of communication for worst-case inputs.

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